Muhammad Muzammal

Software Engineer (.NET) · AI-Native Development · Identity & Secure Healthcare APIs

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Software Engineer with 5 years building secure, HIPAA-regulated healthcare platforms in C#/.NET, SQL Server, Angular and Azure. Leads identity work, from Keycloak SSO and a zero-downtime migration off legacy auth to root-cause analysis of production incidents. Works AI-first: built guardrails and tooling for AI coding agents on Claude Code (custom skills, policy hooks, MCP servers, scheduled agents and an evaluation harness), so AI-generated changes are reviewed, tested and measured before they ship. MCS Gold Medalist.

EXPERIENCE

Software Engineer · iCareManager (US healthcare SaaS)

Nov 2023 – Present · Hybrid

IDENTITY & PLATFORM

- Built a custom **Keycloak Admin SDK** covering user lifecycle, multi-tenancy, MFA/lockout controls, impersonation and audit-tracked force logout.
- Designed **dual-path auth middleware** that moved facilities from legacy auth to Keycloak one at a time with zero downtime.
- Designed event-driven staff sync on **Azure Service Bus**: fan-out to 6 subscribers, MediatR routing, PeekLock delivery, dead-letter/retry handling and a parallel consumer for backlog throughput.
- Hardened the session hand-off by replacing URL tokens with **single-use, server-side nonces** across three application layers.
- Built 2-layer browser session detection (SharedWorker + SignalR) with grace-period logic for accurate tab-close logout.

AI-ASSISTED ENGINEERING · CLAUDE CODE

- Built **guardrails for AI coding agents**: approval gates before production changes, automated review on high-risk code, and blocks on database writes, leaked secrets and known bug patterns.
- Wrote **custom agent skills** for repeatable work such as deployments, incident investigations and pull-request review.
- Set up **MCP servers** giving agents scoped, approval-gated access to code search and browser automation in test environments.
- Built an **LLM evaluation harness** that replays real agent sessions as test cases and compares results on every model or config change, flagging regressions and cost increases.
- Automated recurring work with **scheduled agents**: a Jira watcher that drafts fix proposals for new tickets, a daily Datadog issue digest and auto-generated stand-ups.

PRODUCTION RELIABILITY

- Led root-cause analysis of production incidents with **Datadog APM**, SQL Server Query Store and code tracing: thread-pool starvation from sync-over-async calls, infinite recursion in an ORM merge path, and DB CPU saturation from unindexed queries. Shipped fixes and wrote RCA reports for leadership.
- Removed a 504 in MFA reset by eliminating a per-user loop across large organizations; sped up identity sync with a targeted functional index.

PRODUCT DELIVERY

- Built a Staff Attendance Management System (ASP.NET Core Web API + Angular) with real-time monitoring, automated reporting and HR integration.

SKILLS

AI engineering

`Claude Code` `MCP` `AI agents` `LLM evaluation` `AI code review` `Prompt & context engineering`

Backend

`C#` `.NET 6–10` `ASP.NET Core` `Web API` `EF Core` `MediatR` `Clean Architecture` `DDD` `Quartz.NET`

Identity & security

`Keycloak` `OAuth 2.0` `OpenID Connect` `SSO` `MFA` `OWASP / CWE` `HIPAA`

Cloud & data

`Azure` `Service Bus` `Key Vault` `Dapr` `Docker` `Kubernetes` `SQL Server` `PostgreSQL`

Frontend & quality

`Angular` `AngularJS` `SignalR` `Playwright` `NUnit` `Moq` `Datadog` `Jira`

- Architected a Providers module (registration, credentials, service tracking) on clean architecture. Mentor junior developers and run code reviews.

Associate Software Engineer · iCareManager `Sep 2022 – Nov 2023`

- Full-stack development on ASP.NET Core (.NET 5/6) and Angular 10 for healthcare management systems.
- Built Training Management features (course creation, enrollment, progress tracking) with EF Core and SQL Server.
- Worked on the Electronic Medication Administration Record (eMAR) system with a focus on compliance and patient safety. Unit tests with NUnit and Moq.

Associate Software Engineer · HazelSoft `Feb 2022 – Aug 2022`

- Built a Document Management System with versioning, access permissions and automated workflows (ASP.NET Core Web API).
- Implemented background scheduling with Quartz.NET; designed SQL Server schemas and tuned queries.

Associate Software Engineer · Cyber Space `Sep 2021 – Jan 2022`

- Built multi-provider social sign-in (LinkedIn, Pinterest, Google, TikTok) for the SparesWorld platform.
- Built a Yodlee financial-data API wrapper for Mortgage House Solutions using clean architecture and the Mediator pattern.

EDUCATION

Master of Computer Science
University of Management and Technology (UMT)
2019 – 2021 · CGPA 3.84/4 · **Gold Medalist**

Associate Degree, Computer Science
Government College University, Faisalabad
2016 – 2018 · CGPA 3.84/4

CERTIFICATIONS

DevOps Practices & MLOps
Dice Analytics · 2025 · Docker, Swarm, Kubernetes

C#, SQL, ASP.NET MVC
Aptech Pakistan · 2019